

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = x^2 + 14x + 48$

$x + 8 = 11$

The solution to the given system of equations is (x, y) . What is the value of y ?

Correct Answer: 99

Rationale

The correct answer is **99**. In the given system of equations, the second equation is $x + 8 = 11$. Subtracting **8** from both sides of this equation yields $x = 3$. In the given system of equations, the first equation is $y = x^2 + 14x + 48$. Substituting **3** for x in this equation yields $y = (3)^2 + 14(3) + 48$, or $y = 99$. Therefore, the solution to the given system of equations is $(x, y) = (3, 99)$. Thus, the value of y is **99**.